

3.1.2 Performance characteristics of materials

Performance characteristics of fibres

Content	Potential links to maths and science
Students should be able to name different types of fibres. Students should be able to describe the performance characteristics of fibres, including: • absorbency • strength • elasticity • flammability • thermal qualities • lustre • handle. Students should be able to explain the different fibres and their specific performance characteristics, including: • plant/cellulose fibres: very absorbent, little elasticity, good strength, poor insulators • animal/protein fibres: very absorbent, thermally insulating • regenerated fibres: poor strength, soft, highly absorbent, crease badly • new generation lyocells: with improved performance characteristics • synthetic fibres: very good strength, totally non-absorbent, smooth, lightweight, do not crease, can be heat set (thermoplastic). Students should be able to explain the shape and formation of fibres. They should understand and be able to discuss the different cross-sectional and linear formation of fibres that can occur in natural form, and those that can be engineered during synthetic and manmade fibre production.	Understand how the physical shape and formation of fibres affect their performance.

Content	Potential links to maths and science
Fibre production. Students should be aware of, and be able to describe, the manufacturing processes used to make fibres in relation to the fibre source. With specific reference to staple fibres and continuous filament fibres, students should be able to demonstrate a knowledge of: • wet spinning of regenerated fibres • melt spinning of synthetic fibres to give very fine, smooth continuous filaments.	Understand how fibre production can affect the fibre formation.
Yarn production. Students should be aware of, and be able to describe, the manufacture of staple and filament yarns, single and plied yarns.	
Mixture and blends. Students should know of the need to blend fibres to create aesthetic effects, performance fabrics, improved care and maintenance of fabrics. Students should be aware of, and be able to describe, the production processes associated with mixtures and blends, including: • the mechanics of blending different fibres together • how fibres are combined together to make yarns, eg staple fibre blends and multi-filament yarns • yarns made from mixes of staple fibres and filament fibres cut down to staple form • the different methods of core-spinning yarns which include elastomeric fibres • fibre content of typical blends.	
Hand and machine knitted methods. Students should be aware of, and be able to describe, the production processes associated with hand and machine knitting, including: • panel knitting • fully fashioned panels • whole garment knitting.	

Content	Potential links to maths and science
Non-woven fabrics. Students should be aware of, and be able to describe, the production processes associated with non-woven fabrics, including: • felts, adhesive and heat bonded • needle-felt.	
Special woven effects. Students should be aware of, and be able to describe, the production processes associated with woven effects with coloured yarns, including: • checks and stripes including gingham, tartan, madras.	
Effects created using fibres and yarns. Students should be aware of, and be able to describe, the effects that can be created, including: • bouclé and crêpe fabrics • crinkle and permanent creasing • metallic fibres • use of dyes to give multicoloured fabric • use of thermoplastic fibres.	

Performance characteristics of yarns

Content	Potential links to maths and science
Students should know that fibres need to be made into yarns before they can be manufactured into woven and knitted fabrics. Students should be able to explain the performance characteristics of yarns, including: • the importance of twist in relation to strength and bulk of the yarn: the importance of this in the making of crêpe yarns and fabrics • technical terms relating to yarns: systems for numbering yarns, eg tex and denier staple and filament yarns. Students should be able to describe the main yarn types, including: • textured yarns: false twist and air jet texturing processes, why yarns need to be textured, the importance of thermoplastic qualities in the texturing process • fancy yarns: bouclé, slub, chenille, metallised yarns.	

Performance characteristics of woven fabrics

Content	Potential links to maths and science
Students should be able to explain the main fabric structures of woven fabrics and be able to recognise these structures and typical end uses for a range of woven fabrics, including: • plain (tabby) weave fabrics: broderie anglaise, calico, canvas, chambray, chiffon, flannel, muslin, organdie, poplin, sheeting, shirting, taffeta, voile, winceyette • twill weave fabrics: cavalry twill, denim, dog-tooth check, drill, gabardine, herringbone tweeds, serge, tartan • satin weave fabrics: satin, sateen, duchesse satin, heavy bridal satins, lighter weight satins for linings and lingerie • brocades and Jacquard • three yarn system weaves: cut pile fabrics, including velvet, needlecord, corduroy, loop pile fabrics, including terry towelling.	

Performance characteristics of knitted fabrics

Content	Potential links to maths and science
Students should be able to explain the two main structures of knitted fabrics and be able to recognise these structures and typical end uses for a range of knitted fabrics, including: • weft knits: hand and machine knits, plain knit, single jersey, double jersey, rib knits, jacquard knits • warp knits: net and lace structures.	

Performance characteristics of non-woven fabrics

Content	Potential links to maths and science
Students should be able to understand and explain that non-woven fabrics are produced directly from fibres and have knowledge of typical end uses of non-woven fabrics.	

Smart materials

Content	Potential links to maths and science
Students should know and understand the term smart material. The effects that can be created by a range of smart materials and have knowledge of specific applications, eg in relation to fabrics for safety and novelty products. Students should be able to explain the suitability of smart materials for given applications making reference to how the material responds to external stimuli, including: • changes in temperature • changes in light levels • changes in pressure (force).	

Technical textiles

Content	Potential links to maths and science
Students should know and understand the term technical textiles. Students should be able to explain the suitability of technical textiles for given applications. Students should be familiar with the following technical textiles: • phosphorescent materials, microencapsulated fibres and fabrics • ceramic and carbon fibres used in the production of nano-fibres • ceramic fibres to give a fabric UV protection properties.	

Commercial names of fibres and fabrics

Content	Potential links to maths and science
Students should be able to name and describe popular names of natural, man-made and synthetic fibres and fabrics, including: • Tactel® • Lyocell® • Modal® • Tencel® • Lycra® • Polar fleece.	

Performance characteristics of fabrics

Content	Potential links to maths and science
Students should be able to explain the performance characteristics of fabrics, including: • strength • durability • elasticity • flammability • thermal qualities • absorption • water-resistance • stretch • formability • handle • drape • weight • rip-proof. Students should be able to explain the following terms, and how they relate to particular fabrics: • pattern repeat • directional pile • nap • texture • lustre. Students should be able to describe and explain the following, and how they relate to particular fabrics and their uses: • woven fabrics have relatively good strength and stability • non-woven fabrics lack strength and have no grain • knitted fabrics have fluidity and stretch • a nap, or pile, reflects light in different ways • size of pattern repeat in relation to the appearance of a product.	Understand how the physical structure of fabrics affects performance. Understand that environmental factors can cause potential degradation of fabrics, eg with reference to colour fastness, weakening by sunlight and chlorine, inappropriate care routines.

Content	Potential links to maths and science
Students should know and understand how the properties of fabrics and their physical characteristics relate to their selection for specific fashion, clothing and textile products. This should include reference to both fabric construction and fibre/yarn content and the inter-relationship between them in determining the overall characteristic of a fabric. Properties and physical characteristics to include: • tensile strength • elasticity • resilience • durability • flammability • weight.	